

THE UNITED REPUBLIC OF TANZANIA
PRESIDENT OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DODOMA REGION FORM SIX MOCK EXAMINATIONS

CODE: 131/2

PHYSICS 2

Time; 3 Hours

November, 2023

INSTRUCTIONS

1. This paper consists of **six (6)** questions, each carry 20 marks.
2. Answer any **five (5)** questions.
3. Mathematical tables and non-programmable calculators may be used.
4. Cellular phones and any unauthorized materials are not allowed in examination room.
5. Write your Examination Number on every page of your answer booklet(s).
6. The following information may be useful;-
 - (i) Acceleration due to gravity, $g = 9.8 \text{ Nkg}^{-1}$ or 9.8 ms^{-2}
 - (ii) Permittivity in free space $\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$
 - (iii) Permeability in free space $\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$
 - (iv) Density of air $= 1.2 \text{ kgm}^{-3}$
 - (v) Density of water $= 1000 \text{ kgm}^{-3}$
 - (vi) Mass of electron $= 9 \times 10^{-31} \text{ kg}$
 - (vii) Surface tension of mercury is $4.72 \times 10^{-1} \text{ Nm}^{-1}$
 - (viii) Speed of light $c = 3 \times 10^8 \text{ ms}^{-1}$
 - (ix) Speed of sound in air $= 320 \text{ ms}^{-1}$
 - (x) Electron charge $= 1.6 \times 10^{-19} \text{ C}$
 - (xi) Plank's constant $= 6.63 \times 10^{-34} \text{ Js}$
 - (xii) Avogadro's number $= 6.02 \times 10^{23} \text{ mol}^{-1}$
 - (xiii) Molar gas constant $= 8.31 \text{ Jmol}^{-1} \text{ K}^{-1}$
 - (xiv) Modulus of rigidity for cube $= 8 \times 10^6 \text{ Nm}^{-2}$
 - (xv) Pie $\pi = 3.14$

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1.
 - (a)
 - (i) Give two assumptions made when treating Bernoulli's equation. **(2 marks)**
 - (ii) According to Bernoulli's principle, the pressure of water flowing through the pipe should remain constant but actually it goes on decreasing. Give reason on this observation. **(3 marks)**
 - (b)
 - (i) State the Poiseuille's theorem and deduce its equation. **(2 marks)**
 - (ii) A tank of cross sectional area A has a viscous liquid to a height h_0 above the base. The liquid is allowed to flow out of the container through a horizontal tube which is narrow and long, connected to the base of the tank. Show that the height h of the remaining liquid at any time t in the tank obey the equation $h = h_0 e^{-ct}$ **(5 marks)**
 - (c)
 - (i) A sphere is dropped under gravity through a fluid of viscosity η . Taking average acceleration as half of initial acceleration, show that the time taken to attain terminal velocity is independent of the fluid density. **(4 marks)**
 - (ii) The static pressure in a horizontal pipeline is $4.3 \times 10^4 \text{ Pa}$, the total pressure is $4.7 \times 10^4 \text{ Pa}$ and the area of cross section is 20 cm^2 . If the fluid flowing through the pipeline is considered to be incompressible, non-viscous and has a density of 10 kg m^{-3} , calculate the flow velocity and the volume flow rate of the fluid in the pipeline **(4 marks)**
2.
 - (a)
 - (i) Give the meaning of surface tension in terms of the molecular theory. **(2 marks)**
 - (ii) A spherical drop of mercury of radius 2mm falls to the ground and breaks into 10 droplets of equal size. Calculate the amount of work to be done in the process. **(4 marks)**
 - (b)
 - (i) How does fatigue failure occurs? **(2 marks)**
 - (ii) A cube of sides 20cm has one side fixed. A tangential force equal to the weight of 400kg is applied to the opposite face. Find the shear strain and distance through which the strained side moved. **(5 marks)**
 - (c)
 - (i) The pressure of a gas in a container increases when the gas is heated. Explain this observation in terms of kinetic theory of gases. **(3 marks)**
 - (ii) Find the mean free path and collision frequency for nitrogen molecules at a temperature of 20°C and pressure of 1 atmosphere, assume a molecular diameter of nitrogen is $2 \times 10^{-10} \text{ m}$ and the average speed of nitrogen molecule at 20°C is 511m/s **(4 marks)**

3. (a) (i) Using example in each, explain the terms damped oscillation, forced oscillation and resonance as applied in vibration and waves. **(3 marks)**
- (ii) What is happening during the propagation of electromagnetic waves in a medium? **(2 marks)**
- (b) (i) The value of numerical aperture of the objective lens of a microscope is 1.25. If the light of wavelength $5000\text{ \AA}$ is used, determine the minimum separation between two points to be seen as a distinct. **(3 marks)**
- (ii) In a Young's double slit experiment with slit separation 0.1mm , one observes a bright fringes at an angle $\frac{1}{4}\text{rad}$ by using light of wavelength λ_1 when the light of wavelength λ_2 is used the bright fringe is seen at the same angle in the same set up. If both wavelength are in visible range 380nm to 740nm , calculate the value of λ_1 and λ_2 **(5 marks)**
- (c) (i) Explain the meaning of Doppler effect. **(2 marks)**
- (ii) A horn of frequency 900Hz is sounded by a car travelling towards a cliff and normal to the cliff with a velocity of 20m/s . Calculate the beat frequency of the horn sound as heard by the car driver. **(5 marks)**
4. (a) (i) State Coulomb charges law. **(2 marks)**
- (ii) The point charges are located at x-axis of a coordinated system, $Q_1 = +1.0\text{nC}$ at $x = +2.0\text{cm}$ and $Q_2 = +3.0\text{nC}$ at $x = +4.0\text{cm}$. What is the total force exerted by Q_1 and Q_2 on the third charge $Q_3 = +5\text{nC}$ at $x = 0\text{cm}$ **(4 marks)**
- (b) (i) The region between two concentric spheres of a radii a and b respectively as shown in figure 1 has volume charge density $\rho = \frac{A}{r}$, where A is the a constant. At the centre of the spheres is a point charge Q . Calculate the value of a constant A .

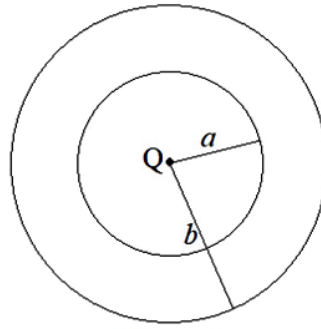
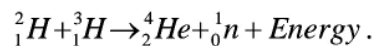


Figure 1

- (ii) Assume that an electric field $E = 30X^2i$ exists in space. Then, the potential difference $V_A - V_0$, where V_0 is the potential at the origin is and V_A is the potential at $X = 2m$. Calculate the difference in potential. **(4 marks)**
- (c) (i) Four charges equal to $-Q$ are placed at four corners of a square and a charge q is at its centre. Calculate the value of q if the system is in equilibrium. **(4 marks)**
- (ii) A sheet of aluminium foil of negligible thickness is introduced between the plates of a capacitor. Determine the capacitance of the capacitor. **(2 marks)**
5. (a) (i) Give one difference and one similarity between the electric force and magnetic force on a charged particle **(2 marks)**
- (ii) A rectangular coil of dimensions 5cm x 2.5cm with 100 turns, carrying a current of 3A in the clockwise direction is kept centered at the origin and in the X-Z plane. A magnetic field of 1.0 T is applied along X-axis determine the torque on the coil. **(3 marks)**
- (b) (i) State the Biot-Savart law. **(2 marks)**
- (ii) A thin ring of 10cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of 40π rad/s about its axis perpendicular to its plane. If the magnetic field at its centre is $3.8 \times 10^{-9} T$, calculate the charge carried by the ring. **(4 marks)**
- (c) (i) Briefly explain the magnetic properties exhibited by ferromagnetic materials. Give five points. **(5 marks)**
- (ii) The magnetic field due to a current carrying circular loop of radius 3cm at a point on the axis at a distance of 4cm from the centre is $54\mu T$. what will be its value at the centre of the loop? **(3 marks)**

6. (a) (i) The hydrogen atom spectrum consists of a large number of spectral lines and yet this atom has only one electron. Give explanation on this observation. **(3 marks)**
- (ii) If the electron in the hydrogen has energy level of -13.6eV , calculate the maximum wavelength produced in Balmer series of the hydrogen atom. **(4 marks)**
- (b) The maximum energy of photoelectrons emitted from a metal plate is 1.2eV . If the threshold wavelength is $2.48 \times 10^{-7}\text{ m}$, calculate the frequency of the incident light. **(6 marks)**
- (c) (i) Distinguish between nuclear fusion and nuclear fission. **(2 marks)**
- (ii) The energy in the interior of the sun is produced by fusing deuterium and tritium to form Helium and neutron as summarized in the nuclear equation.



How much ${}^2_1\text{H}$ in kg/s is required to generate 100 MW by fusion? Some atomic and particles masses may be used; ${}^4_2\text{He} = 4.002603u$, ${}^2_1\text{H} = 2.014102u$, ${}^3_1\text{H} = 3.016049u$ and ${}^1_0\text{n} = 1.008665u$. **(5 marks)**